

实验 0 《复杂网络与大数据分析》环境的搭建

一、实验目的

《复杂网络与大数据分析》涉及到复杂网络的构建、网络基本拓扑参数计算、复杂网络同配性分析、静态网络社团检测算法实现、网络节点重要性排序、随机网络模型构建及拓扑分析、小世界网络模型构建及拓扑分析、无标度网络模型构建及拓扑分析等内容。本课程采用 python 语言对复杂网络以上内容进行实现，需要安装一系列软件：anaconda+Jupyter notebook 或者 PyCharm 或者 spyder，以及相关的软件包 networkx, numpy, matplotlib 等。本实验的主要目的是安装所需软件、并熟悉其用法，为后续实验做好准备。

二、实验内容

1、安装 anaconda 包管理工具

(1) 参考 anaconda 的安装.docx 文档进行安装。

(2) 创建环境。

运行 anaconda，如图：



输入命令 `conda create -n Mynetwork python=3.9`，点击回车。

其中 Mynetwork 是环境的名字，可以自己定义，`python=3.9` 是这个环境将使用 3.9 的 python 版本，大家可以根据自己 python 的版本进行环境创建，也可以省略红色部分。

过程如图：

```

Anaconda Prompt (anaconda) - conda create -n Mynetwork python=3.9
Total: 25.7 MB

The following NEW packages will be INSTALLED:

bzip2                conda-forge/win-64::bzip2-1.0.8-h8ffe710_4 None
ca-certificates      conda-forge/win-64::ca-certificates-2022.12.7-h5b45459_0 None
libffi               conda-forge/win-64::libffi-3.4.2-h8ffe710_5 None
libsqlite             conda-forge/win-64::libsqlite-3.40.0-hcfcfb64_0 None
libzlib              conda-forge/win-64::libzlib-1.2.13-hcfcfb64_4 None
openssl              conda-forge/win-64::openssl-3.0.8-hcfcfb64_0 None
pip                  conda-forge/noarch::pip-23.0-pyhd8ed1ab_0 None
python               conda-forge/win-64::python-3.9.16-h4de0772_0.cpython None
setuptools            conda-forge/noarch::setuptools-67.1.0-pyhd8ed1ab_0 None
tk                   conda-forge/win-64::tk-8.6.12-h8ffe710_0 None
tzdata               conda-forge/noarch::tzdata-2022g-h191b570_0 None
ucrt                  conda-forge/win-64::ucrt-10.0.22621.0-h57923b3_0 None
vc                    conda-forge/win-64::vc-14.3-hb6edc58_10 None
vs2015_runtime        conda-forge/win-64::vs2015_runtime-14.34.31931-h4c5c07a_10 None
wheel                 conda-forge/noarch::wheel-0.38.4-pyhd8ed1ab_0 None
xz                    conda-forge/win-64::xz-5.2.6-h5d14728_0 None

Proceed ([y]/n)? y

Downloading and Extracting Packages
pip-23.0                1.3 MB ##### 100%
vc-14.3                 15 KB ##### 100%
vs2015_runtime-14.34    709 KB ##### 100%
python-3.9.16          16.3 MB ##### 31%

```

运行后如图：

```

Anaconda Prompt (anaconda)
done
#
# To activate this environment, use
#
#     $ conda activate Mynetwork
#
# To deactivate an active environment, use
#
#     $ conda deactivate
#
Retrieving notices: ...working... done
(base) C:\Users\Ding>

```

环境创建后：

此电脑 > 新加卷 (D:) > anaconda > envs				
名称	修改日期	类型	大小	
bioinformatics	2022/12/22 10:39	文件夹		
dgl	2022/12/22 9:16	文件夹		
dingbio	2022/12/23 10:30	文件夹		
Mynetwork	2023/2/8 15:33	文件夹		
pyTorchEnv	2022/7/29 15:22	文件夹		
.conda_envs_dir_test	2022/4/20 8:17	CONDA_ENVS_D...	0 KB	

激活环境：

```

Anaconda Prompt (anaconda)
done
#
# To activate this environment, use
#
#     $ conda activate Mynetwork
#
# To deactivate an active environment, use
#
#     $ conda deactivate
#
Retrieving notices: ...working... done
(base) C:\Users\Ding>conda activate Mynetwork

```

激活后

```
Anaconda Prompt (anaconda)
done
# To activate this environment, use
# $ conda activate Mynetwork
# To deactivate an active environment, use
# $ conda deactivate
Retrieving notices: ...working... done
(base) C:\Users\Ding>conda activate Mynetwork
(Mynetwork) C:\Users\Ding>_
```

删除环境（有需要的话，谨慎）

键入

```
conda env remove -n Mynetwork
```

2、安装 Jupyter notebook 及各种依赖包

在上图的窗口键入

```
conda install jupyter notebook
```

```
Anaconda Prompt (anaconda) - conda install jupyter notebook
done
# To activate this environment, use
# $ conda activate Mynetwork
# To deactivate an active environment, use
# $ conda deactivate
Retrieving notices: ...working... done
(base) C:\Users\Ding>conda activate Mynetwork
(Mynetwork) C:\Users\Ding>conda install jupyter notebook
Collecting package metadata (current_repodata.json): -
```

键入

```
conda install numpy matplotlib pip pandas
```

```
Anaconda Prompt (anaconda) - conda install numpy matplotlib pip pandas

libxml2 conda-forge/win-64::libxml2-2.10.3-hc3477c8_0 None
m2w64-gcc-libgfortran conda-forge/win-64::m2w64-gcc-libgfortran-5.3.0-6 None
m2w64-gcc-libs conda-forge/win-64::m2w64-gcc-libs-5.3.0-7 None
m2w64-gcc-libs-core conda-forge/win-64::m2w64-gcc-libs-core-5.3.0-7 None
m2w64-gmp conda-forge/win-64::m2w64-gmp-6.1.0-2 None
m2w64-libwinpthread conda-forge/win-64::m2w64-libwinpthread-git-5.0.0.4634.697f757-2 None
matplotlib conda-forge/win-64::matplotlib-3.6.3-py39hcbf5309_0 None
matplotlib-base conda-forge/win-64::matplotlib-base-3.6.3-py39haf65ace_0 None
mkl conda-forge/win-64::mkl-2022.1.0-h6a75c08_874 None
msys2-conda-epoch conda-forge/win-64::msys2-conda-epoch-20160418-1 None
munkres conda-forge/noarch::munkres-1.1.4-pyh9f0ad1d_0 None
numpy conda-forge/win-64::numpy-1.24.2-py39h16ffa76_0 None
openjpeg conda-forge/win-64::openjpeg-2.5.0-ha2aaf27_2 None
pandas conda-forge/win-64::pandas-1.5.3-py39h2ba5b7c_0 None
pillow conda-forge/win-64::pillow-9.4.0-py39hcebd2be_1 None
pthread-stubs conda-forge/win-64::pthread-stubs-0.4-hcd874cb_1001 None
pthreads-win32 conda-forge/win-64::pthreads-win32-2.9.1-hfa6e2cd_3 None
pyparsing conda-forge/noarch::pyparsing-3.0.9-pyhd8ed1ab_0 None
pytz conda-forge/noarch::pytz-2022.7.1-pyhd8ed1ab_0 None
tbb conda-forge/win-64::tbb-2021.7.0-h91493d7_1 None
unicodedata2 conda-forge/win-64::unicodedata2-15.0.0-py39ha55989b_0 None
xorg-libxau conda-forge/win-64::xorg-libxau-1.0.9-hcd874cb_0 None
xorg-libxdmcp conda-forge/win-64::xorg-libxdmcp-1.1.3-hcd874cb_0 None

Proceed ([y]/n)? y

Downloading and Extracting Packages
pandas-1.5.3 | 10.3 MB | #####9 41%
```

键入：conda install networkx

```
Anaconda Prompt (anaconda) - conda install networkx

(base) C:\Users\Ding>conda activate Mynetwork
(Mynetwork) C:\Users\Ding>conda install networkx
Collecting package metadata (current_repodata.json): done
Solving environment: /
```

注：以上软件包可以同时安装

3、将虚拟环境 Mynetwork 加载到 jupyter notebook 的 kernel

键入

conda install -n Mynetwork ipykernel

由于我的已经安装，出现下面的界面

```
Anaconda Prompt (anaconda) - conda install -n Mynetwork ipykernel

(base) C:\Users\Ding>conda activate Mynetwork
(Mynetwork) C:\Users\Ding>conda install -n Mynetwork ipykernel
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 22.9.0
  latest version: 22.11.1

Please update conda by running

  $ conda update -n base -c conda-forge conda

# All requested packages already installed.
Retrieving notices: ...working... done
(Mynetwork) C:\Users\Ding>
```

键入

activate Mynetwork

键入

python -m ipykernel install --user --name Mynetwork --display-name Mynetwork

```

Anaconda Prompt (anaconda) - conda install jupyter notebook - conda install numpy matplotlib pip pandas - conda install -n MyNetwor...
(MyNetwork) C:\Users\Ding>python -m ipykernel install --user --name Mynetwork --display-name Mynetwork
Installed kernelspec Mynetwork in C:\Users\Ding\AppData\Roaming\jupyter\kernels\mynetwork
(MyNetwork) C:\Users\Ding>
```

4、运行 jupyter notebook

```

Anaconda Prompt (anaconda) - conda install jupyter notebook - conda install numpy matplotlib pip pandas - conda install -n MyNetwor...
(MyNetwork) C:\Users\Ding>python -m ipykernel install --user --name Mynetwork --display-name Mynetwork
Installed kernelspec Mynetwork in C:\Users\Ding\AppData\Roaming\jupyter\kernels\mynetwork
(MyNetwork) C:\Users\Ding>jupyter notebook
[I 16:48:40.730 NotebookApp] Serving notebooks from local directory: C:\Users\Ding
[I 16:48:40.730 NotebookApp] Jupyter Notebook 6.5.2 is running at:
[I 16:48:40.730 NotebookApp] http://localhost:8888/?token=ff3ae5684810d6701e0b0b4da6f02c84e82efffb47a3453
[I 16:48:40.730 NotebookApp] or http://127.0.0.1:8888/?token=ff3ae5684810d6701e0b0b4da6f02c84e82efffb47a3453
[I 16:48:40.730 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
C 16:48:40.862 NotebookApp]

To access the notebook, open this file in a browser:
file:///C:/Users/Ding/AppData/Roaming/jupyter/runtime/nbserver-19432-open.html
Or copy and paste one of these URLs:
http://localhost:8888/?token=ff3ae5684810d6701e0b0b4da6f02c84e82efffb47a3453
or http://127.0.0.1:8888/?token=ff3ae5684810d6701e0b0b4da6f02c84e82efffb47a3453
```

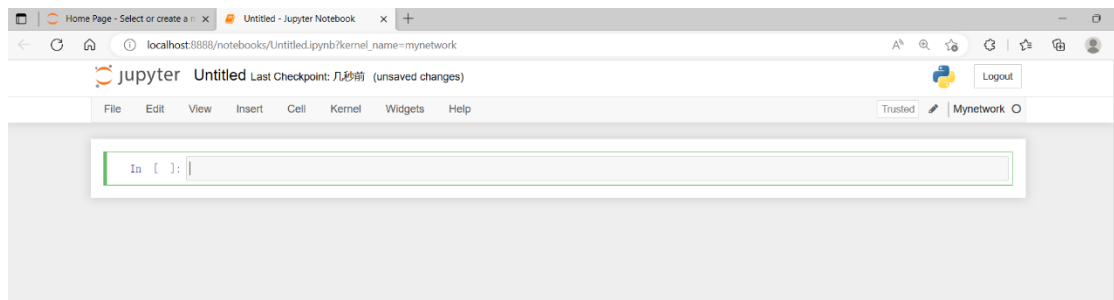
可以看到如下界面：



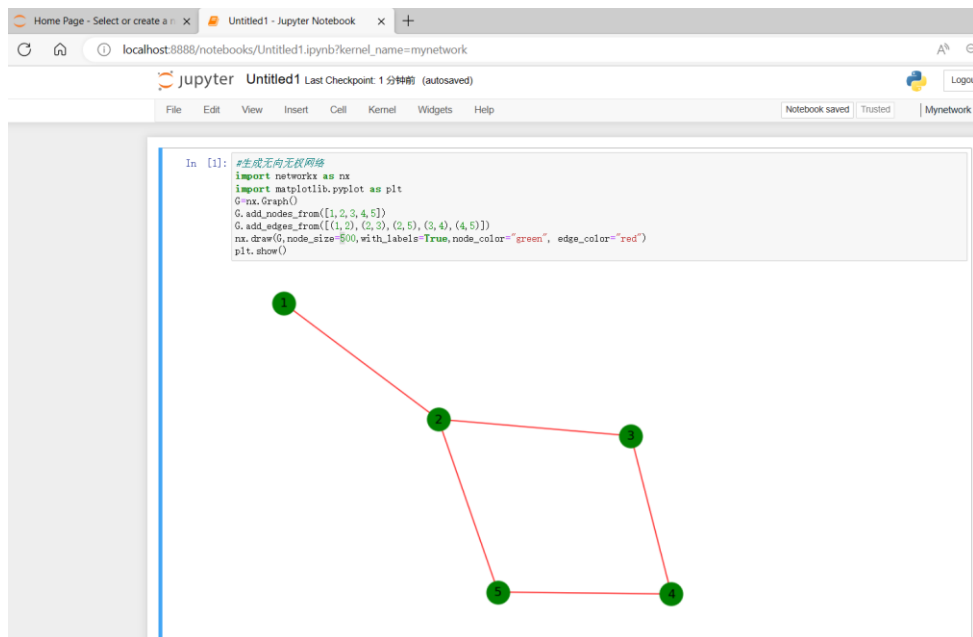
其中蓝色框就是刚刚创建的 anaconda 的虚拟环境。

点击 Mynetwork，出现新的窗口，可以在其中进行编程和文档编辑，jupyter notebook 的

使用可以参考相关教程，如 <https://docs.jupyter.org/en/latest/>。



例如键入如下程序运行后：



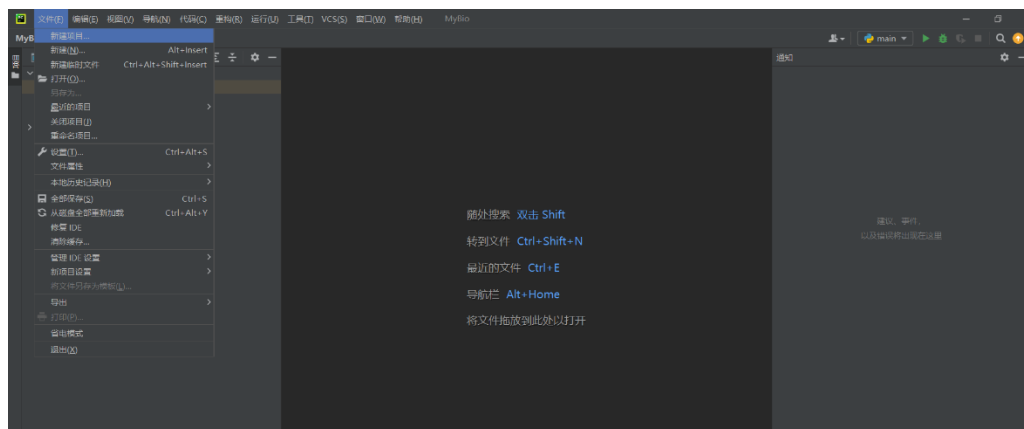
附：

参考网页：<https://blog.csdn.net/miracleoa/article/details/106115730>

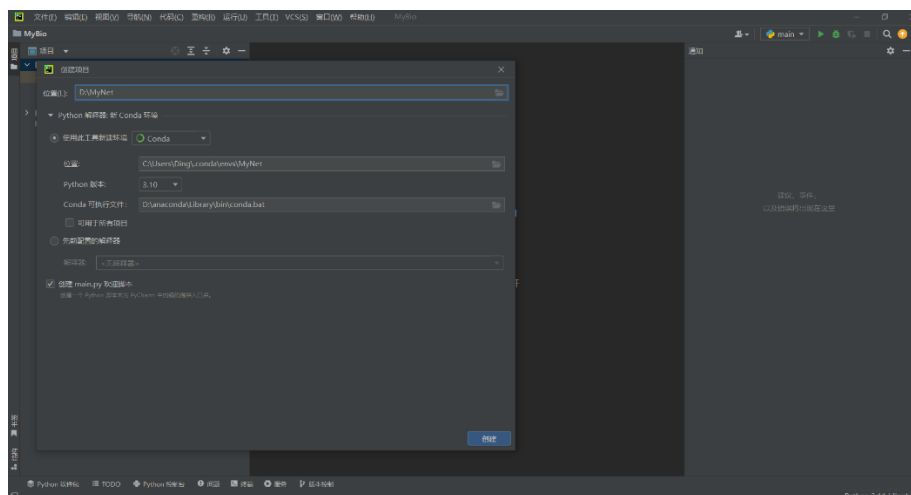
5、pycharm 软件安装

进官网：<https://www.jetbrains.com/pycharm/download/>

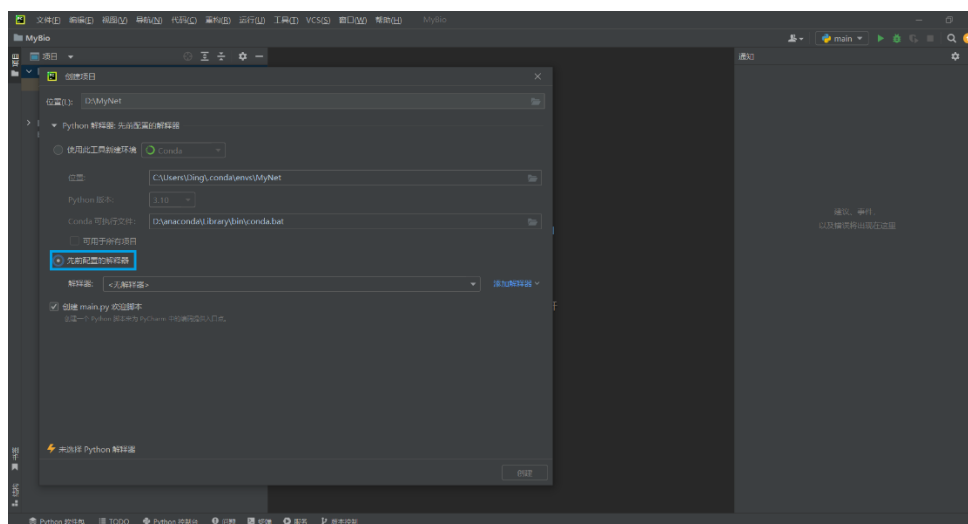
下载 community 版本，双击.exe 文件安装，安装后运行，点击新建项目，如下图：



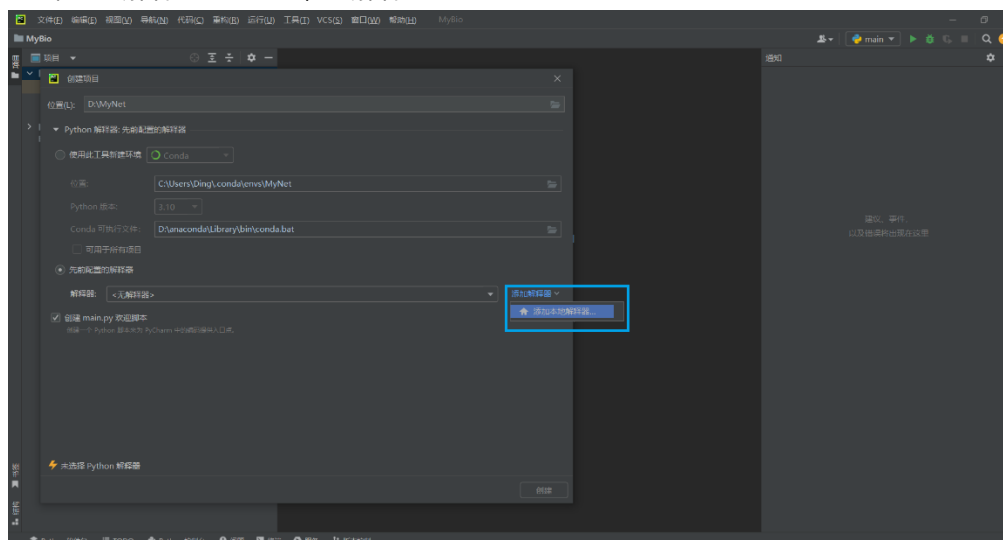
在下图的位置处键入存放项目的路径和项目名称



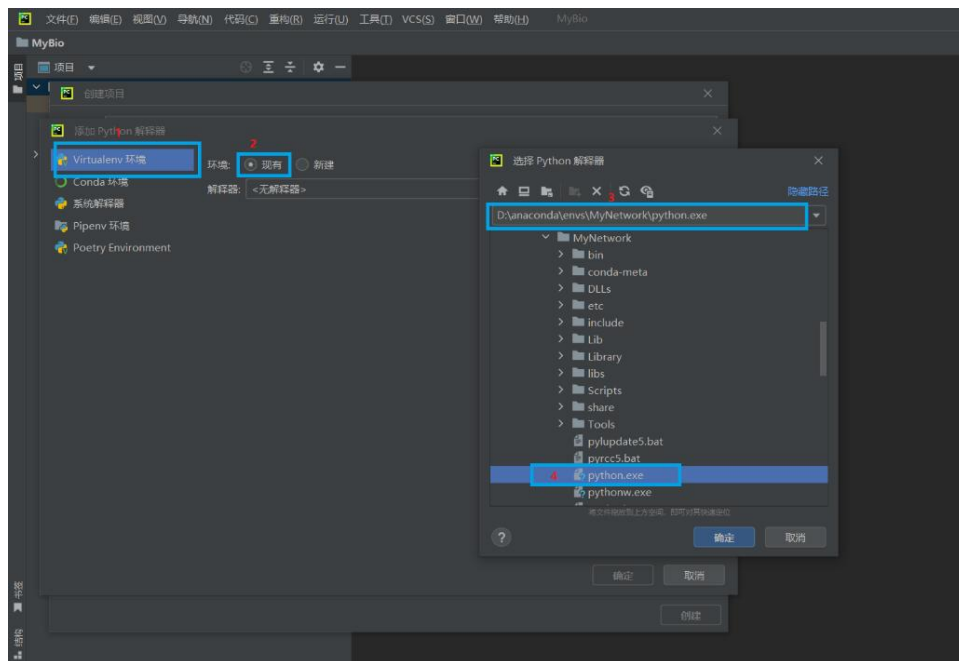
选先前配置的解器器，如下图蓝框



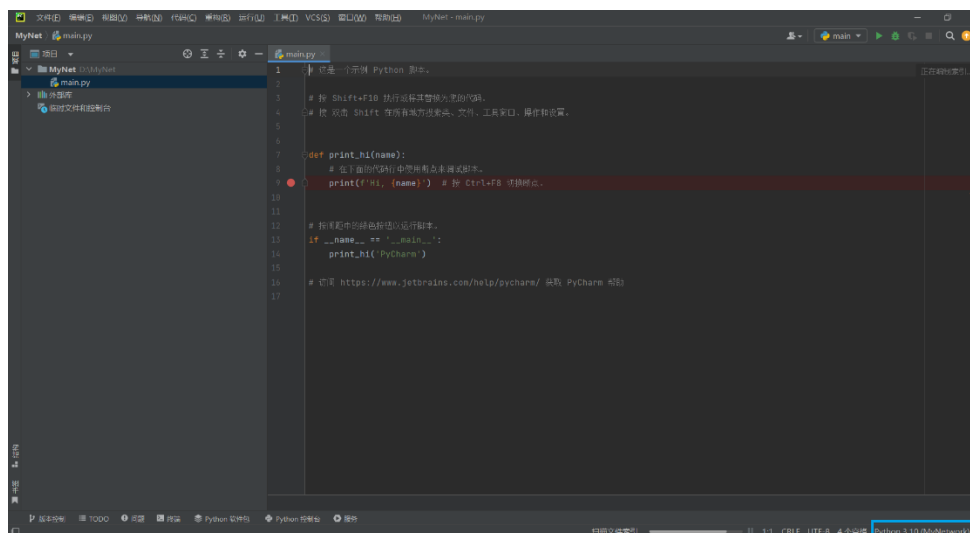
点击添加解释器->添加本地解释器



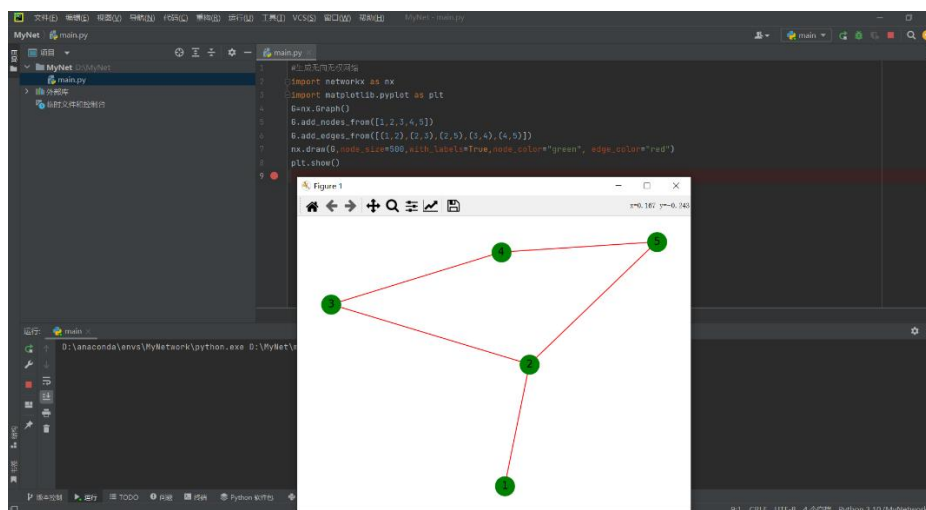
按照下面的数字进行选择和设置：



确定->确定->创建后:

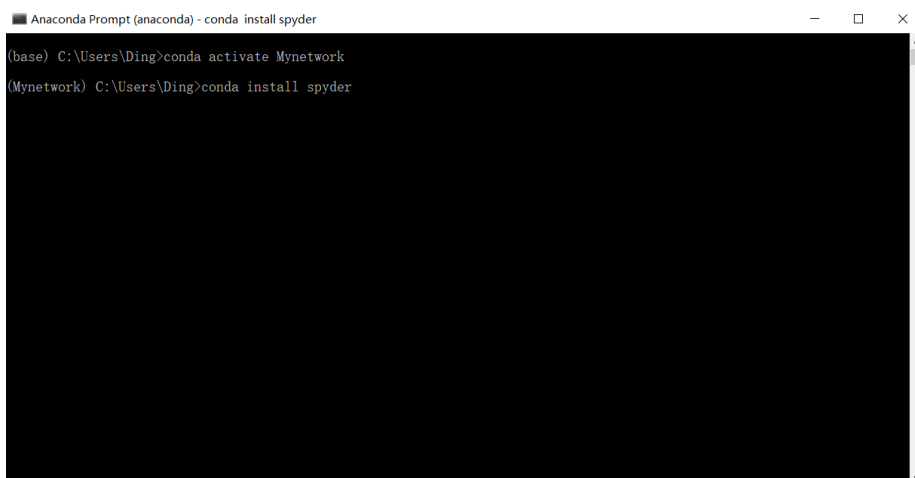


注意此时上图右下角的解释器就是之前的 Mynetwork 虚拟环境中的 python 解释器
在 pycharm 中运行刚刚在 jupyter notebook 中运行的程序



6、spyder 的安装

在 Mynetwork 虚拟环境下，键入
conda install spyder

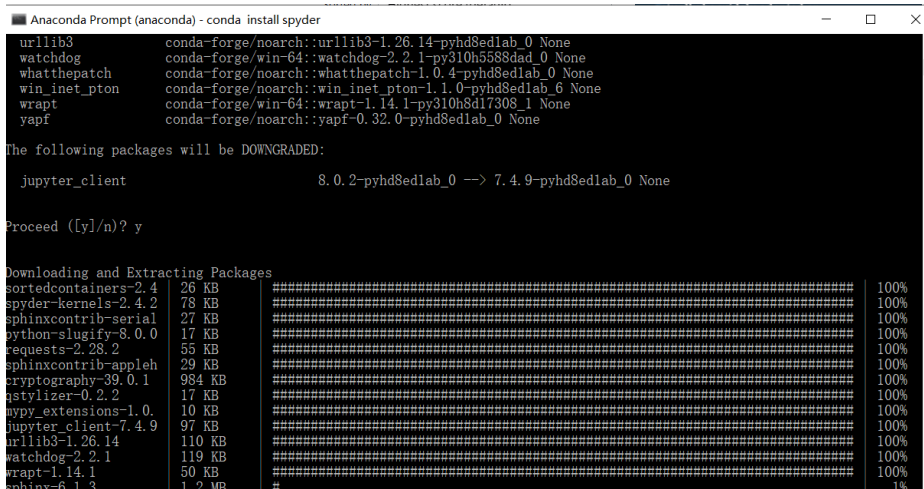


```

(base) C:\Users\Ding>conda activate Mynetwork
(Mynetwork) C:\Users\Ding>conda install spyder

```

过程如下：



```

urllib3 conda-forge/noarch::urllib3-1.26.14-pyhd8ed1ab_0 None
watchdog conda-forge/win-64::watchdog-2.2.1-py310h5588dad_0 None
whatthepatch conda-forge/noarch::whatthepatch-1.0.4-pyhd8ed1ab_0 None
win_inet_pton conda-forge/noarch::win_inet_pton-1.1.0-pyhd8ed1ab_6 None
wrap conda-forge/win-64::wrap-1.14.1-py310h8d17308_1 None
yapf conda-forge/noarch::yapf-0.32.0-pyhd8ed1ab_0 None

The following packages will be DOWNGRADED:
  jupyter_client 8.0.2-pyhd8ed1ab_0 --> 7.4.9-pyhd8ed1ab_0 None

Proceed ([y]/n)? y

Downloading and Extracting Packages
sortedcontainers-2.4 26 KB ##### 100%
spyder-kernels-2.4.2 78 KB ##### 100%
sphinxcontrib-serial 27 KB ##### 100%
python-slugify-8.0.0 17 KB ##### 100%
requests-2.28.2 55 KB ##### 100%
sphinxcontrib-apple 29 KB ##### 100%
cryptography-39.0.1 984 KB ##### 100%
astyle-0.2.2 17 KB ##### 100%
npy_extensions-1.0 10 KB ##### 100%
jupyter_client-7.4.9 97 KB ##### 100%
urllib3-1.26.14 110 KB ##### 100%
watchdog-2.2.1 119 KB ##### 100%
wrap-1.14.1 50 KB ##### 100%
sphinx-6.1.3 1.2 MB ##### 1%

```

在 spyder 中运行刚刚在 jupyter notebook 中运行的程序

